

RAG Evaluation Report

Comparative evaluation of llama3.2:latest and qwen2.5:1.5b

Run	20260728T113753Z	Dataset	1 (40 result rows)
Judge enabled	True	Cosine threshold	0.75 (provisional, not calibrated)

Important interpretation note. No combined score is calculated. Cosine similarity measures semantic agreement with the reference answer, but it does not prove factual correctness.

Per-model metrics

Answer quality and judge scores

Model	N	Cosine mean	Cosine median	Cosine pass	Judge correct	Judge complete	Judge faithful	Judge pass
llama3.2:latest	20	0.897	0.913	1.000	3.100	2.900	3.450	0.600
qwen2.5:1.5b	20	0.792	0.744	0.500	1.750	1.800	1.950	0.400

Retrieval coverage and runtime

Model	Hit@5	Recall@5	Hit@10	Recall@10	Gen ms	Errors
llama3.2:latest	0.950	0.950	0.950	0.950	4491.469	0
qwen2.5:1.5b	0.950	0.950	0.950	0.950	2656.809	0

How to read these values. Judge dimensions are scored from 0 to 4. Retrieval hit and recall indicate whether gold evidence appeared among the retrieved passages. Generation latency is reported as mean milliseconds per answer.

Question-level diagnostics

Question	Model	Diagnostics
csf-gv-sc-05-contracts	llama3.2:latest	evidence retrieved; answer incorrect; missing gold claim
sp1299-backup-actions	llama3.2:latest	retrieval miss; missing gold claim; unsupported claim
sp1299-incident-data-preservation	llama3.2:latest	evidence retrieved; answer incorrect; missing gold claim; contradiction
sp80053-ac-7-invalid-logons	llama3.2:latest	evidence retrieved; answer incorrect; missing gold claim
sp80053-at-2-training-timing	llama3.2:latest	evidence retrieved; answer incorrect; missing gold claim
sp80053-cp-9-backup-information	llama3.2:latest	evidence retrieved; answer incorrect; missing gold claim
sp80053-ir-6-incident-reporting	llama3.2:latest	evidence retrieved; answer incorrect; missing gold claim; unsupported claim
owasp-a01-deny-default	llama3.2:latest	missing gold claim
csf-tier-names	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim; unsupported claim; contradiction
csf-gv-sc-05-contracts	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim; unsupported claim; contradiction
sp1299-backup-actions	qwen2.5:1.5b	retrieval miss; missing gold claim; unsupported claim; contradiction
sp1299-incident-data-preservation	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim; unsupported claim
sp80053-at-2-training-timing	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim; unsupported claim
sp80053-au-11-record-retention	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim; unsupported claim
sp80053-cp-9-backup-information	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim
sp80053-ia-5-authenticator-changes	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim; unsupported claim; contradiction
sp80053-si-2-flaw-remediation	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim
owasp-a01-cwe-incidence	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim; contradiction
owasp-a01-deny-default	qwen2.5:1.5b	evidence retrieved; answer incorrect; missing gold claim; unsupported claim
owasp-a10-ssrf-allowlist	qwen2.5:1.5b	unsupported claim

Interpretation

- **Cosine similarity** measures semantic agreement with the reference answer; the 0.75 threshold is provisional.
- **Evidence hit and recall** measure whether gold passages appeared in retrieval, not whether the answer used them correctly.
- **The LLM judge** is a consistent rubric-based aid, not a substitute for expert human review.

Final results

Metric	Llama 3.2	Qwen 2.5 1.5B
Questions	20	20
Mean cosine similarity	0.897	0.792
Cosine pass rate	100%	50%
Judge correctness, 0-4	3.10	1.75
Judge completeness, 0-4	2.90	1.80
Judge faithfulness, 0-4	3.45	1.95
Judge pass rate	60%	40%
Evidence hit/recall at 5	95%	95%
Mean generation time	4.49 s	2.66 s
Technical errors	0	0

Conclusion. llama3.2:latest produced substantially better answers, while qwen2.5:1.5b was about 1.8 seconds faster on average.

Both models had identical retrieval results because the questions were standalone and used the same retriever. The shared retrieval miss was sp1299-backup-actions; most other weak answers had the correct evidence available, identifying **generation - not retrieval - as the primary failure source**.

The three techniques in simple words

1. Cosine similarity

It compares the meaning of the generated answer with the correct answer using local embeddings. Higher is more similar. The 0.75 threshold is provisional; cosine alone cannot reliably catch wrong numbers, negation, or changes such as MUST versus SHOULD.

2. Retrieval evidence coverage

It checks whether the retriever found the exact document evidence needed for the answer. hit@5 means the evidence reached the model; hit@10 shows whether it was retrieved but ranked too low.

3. LLM judge

Groq openai/gpt-oss-120b reads the answer, reference, required claims, evidence, and retrieved context. It scores correctness, completeness, faithfulness, and relevance from 0 to 4, while listing missing, unsupported, or contradictory claims. Candidate model identity is hidden.

Generated from evaluation run 20260728T113753Z. Results should be interpreted alongside human review and future threshold calibration.